INDIVIDUAL PROJECT REPORT

ON

“Arya College ClubSphere: Centralized Club and Event Management Platform”

INDIVIDUAL CONTRIBUTION REPORT — CLUB MANAGEMENT

Submitted in partial fulfillment of the requirement for the award of Degree of

Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

Submitted by

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Individual responsibility: Club Management

PBL Team ID: PBL2627-CSE-A-024

Under the guidance of

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Session: 2026–27

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CERTIFICATE

This is to certify that the individual project report entitled “Arya College ClubSphere — Club Management” has been prepared by Avanshi Kothari (24EARCS031), a student of Bachelor of Technology in Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, affiliated with Rajasthan Technical University, Kota, during the academic session 2026–27.

The report records the student’s individual contribution to the group project Arya College ClubSphere (PBL2627-CSE-A-024), carried out under the guidance of Er. Ram Babu Buri. The contribution described here forms part of a four-member team project; the work of the other team members is acknowledged and is not claimed in this report.

Role	Name	Signature
Project Guide	Er. Ram Babu Buri Dept. of CSE, ACEIT, Jaipur	
Head of Department	Dr. Akhil Pandey Dept. of CSE, ACEIT, Jaipur	
Student	Avanshi Kothari (24EARCS031)	

Date: _____ Place: Jaipur

DECLARATION

I, Avanshi Kothari (24EARCS031), a student of B.Tech. Computer Science & Engineering at Arya College of Engineering & I.T., Jaipur, hereby declare that this individual report describes my own contribution to the group project Arya College ClubSphere, carried out during the session 2026–27 under the guidance of Er. Ram Babu Buri.

The technical content of this report was prepared from the project source code, database schema and module documentation. Features built by teammates are stated as such. The report does not claim sole authorship of ClubSphere or measured production performance.

To the best of my knowledge, this report has not been submitted to any other university or institution for the award of any degree or diploma.

Avanshi Kothari

Enrollment No.: 24EARCS031

B.Tech. (CSE), Session 2026–27

Signature: _____ Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, Er. Ram Babu Buri, for his guidance throughout the planning and development of ClubSphere. His emphasis on clear requirements and honest reporting of status shaped the way I designed and documented the Club Management module.

I am thankful to Dr. Akhil Pandey, Head of the Department of Computer Science & Engineering, Arya College of Engineering & I.T., for the academic environment, and to the faculty of the department for their support.

I am grateful to my teammates — Avantika Mishra, Akansha Sharma and Aditi Maheshwari. My module depends on a shared MySQL schema, AuthenticationFilter and JSP chrome. Working with Git taught me integration and communication within a team.

Finally, I thank my family and friends for their encouragement and patience during the project.

Avanshi Kothari

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ABSTRACT

This report documents my contribution to ClubSphere: Club Management. I implemented the public club directory (search and category), administrative create/edit/status of clubs, assignment of a club head, and club-head membership add/remove. Events, tasks and announcements all foreign-key to clubs, so this module is the shared catalogue of the 26 official Arya societies seeded in the database.

Keywords: Club directory, membership, JDBC, JSP, RBAC

TABLE OF CONTENTS

Certificate
Declaration
Acknowledgement
Abstract
CHAPTER 1 — INTRODUCTION
CHAPTER 2 — PROJECT OVERVIEW AND INDIVIDUAL RESPONSIBILITIES
CHAPTER 3 — TECHNOLOGY STACK
CHAPTER 4 — REQUIREMENTS OF MY MODULE
CHAPTER 5 — DESIGN AND IMPLEMENTATION
CHAPTER 6 — TESTING AND VERIFICATION
CHAPTER 7 — CHALLENGES AND LEARNING OUTCOMES
CHAPTER 8 — CONCLUSION AND FUTURE SCOPE
REFERENCES

CHAPTER 1 — INTRODUCTION

1.1 Background

ClubSphere is the group PBL project (PBL2627-CSE-A-024) that centralises clubs, events, QR registration, tasks and notifications for Arya College of Engineering & I.T. The team report describes the whole system. This individual report documents only my contribution: **Club Management**.

1.2 Purpose of this Report

This report separates my work from that of my teammates, explains the design of my module, how it was tested, and what remains incomplete.

1.3 Problem Addressed by My Module

Students could not browse official clubs in one place, and club heads had no shared roster. A catalogue with admin governance and per-club membership was required.

1.4 Objectives of My Contribution

- Present 26 Arya clubs for public discovery.
- Let admins create, update and deactivate clubs and assign a head.
- Let club heads maintain only their assigned roster.
- Prevent duplicate memberships with a unique (club_id, user_id) key.

CHAPTER 2 — PROJECT OVERVIEW AND INDIVIDUAL RESPONSIBILITIES

2.1 Overall Project Context

ClubSphere is developed by four students and hosted at github.com/Avantikamishra862/Arya_Clubsphere. It is a Java 17 / JSP / Servlet / MySQL application deployed as a Tomcat WAR.

2.2 Module Ownership

Member	Enrollment	Module
Avantika Mishra	24EARCS032	Authentication & User Management
Avanshi Kothari	24EARCS031	Club Management
Akansha Sharma	24EARCS012	Event Management & QR Code Registration
Aditi Maheshwari	24EARCS007	Dashboards, Tasks & Notifications

My owned area is highlighted in the implementation chapter. Shared files (schema.sql, AuthenticationFilter, navbar.jsp, pom.xml) were coordinated with the team.

2.3 In-scope and Out-of-scope

In scope: ClubServlet, ClubService, ClubDAO, ClubMemberDAO, admin club forms, club-head members views, clubs table seed data.

Out of scope: Event QR codes, task assignment UI, authentication hashing — those belong to teammates.

CHAPTER 3 — TECHNOLOGY STACK

I used the shared ClubSphere stack. My module specifically depends on the items below.

Layer	Technology	How I used it
Web	ClubServlet, AdminServlet, ClubHeadServlet	Public and authorised club URLs
Data	clubs, club_members	Head FK, unique membership
UI	explore.jsp, view.jsp, admin club-form	Search, category, roster
Security	AuthenticationFilter prefixes	Writes require Admin or Club Head

CHAPTER 4 — REQUIREMENTS OF MY MODULE

4.1 Users

Visitors browse clubs without login. Admins manage the catalogue. Club heads manage members of one club.

4.2 Functional Requirements

ID	Requirement
FR-C1	Search clubs by keyword and category
FR-C2	View club profile, advisor and members
FR-C3	Admin CRUD and ACTIVE/INACTIVE
FR-C4	Assign head_user_id
FR-C5	Club-head add/remove member

4.3 Non-functional requirements

Public pages remain usable without login. Write operations stay behind RBAC. Deactivation is preferred to hard delete in operations.

CHAPTER 5 — DESIGN AND IMPLEMENTATION

5.1 Request flow

Public GET /clubs → ClubService.searchClubs → ClubDAO. Admin POST /admin/club/save persists club fields. Club-head membership uses unique (club, user).

5.2 Principal classes and views

Layer	Items
Controllers	ClubServlet, AdminServlet club paths, ClubHeadServlet member paths
Service/DAO	ClubService, ClubDAO, ClubMemberDAO
Model	Club, ClubMember
Views	clubs/explore.jsp, clubs/view.jsp, admin/clubs.jsp, club-head/members.jsp

5.3 Implementation notes

Seed data includes technical, cultural, literary, sports, NSS and other societies with faculty advisors. Club-head screens abort if no club is assigned. Cascades exist in SQL; operational practice is to deactivate.

UML figures for the whole system are in the team report and in docs/UML/Arya_College_ClubSphere_UML_Diagrams.pdf.

CHAPTER 6 — TESTING AND VERIFICATION

6.1 Strategy

Manual: search/filter, admin create/toggle, add/remove member, student denied on /admin/clubs. Duplicate membership hits the unique key.

6.2 Representative checks

Check	Result / note
Explore 26 clubs	Seeded directory visible
Admin toggle status	Status updates
Duplicate member	Rejected by unique key
Student /admin/clubs	Access denied

CHAPTER 7 — CHALLENGES AND LEARNING OUTCOMES

7.1 Challenges

Keeping public read access while protecting writes; wiring head_user_id so login can set managedClubId for the club-head dashboard.

7.2 Learning outcomes

I learned how a catalogue table becomes the parent of several modules, and how unique constraints enforce real-world “one membership row” rules.

CHAPTER 8 — CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Club Management gives ClubSphere a stable campus directory. Other modules attach events and work items to these club records.

8.2 Future work in my module

Student self-join requests with approval, club logos/uploads, and richer category analytics.

REFERENCES

[1] Jakarta Servlet 6.0 Specification.

[2] Apache Tomcat 10.1 Documentation.

[3] MySQL 8.0 Reference Manual.

[4] BCrypt password hashing documentation.

[5] ZXing QR code library documentation.

[6] Arya College ClubSphere team project report, Dept. of CSE, ACEIT, Jaipur, Session 2026–27, PBL2627-CSE-A-024.

[7] https://github.com/Avantikamishra862/Arya_Clubsphere